

Bringing Visual(-inertial) Odometries to the Limits: In Search of Robust and Accurate State Estimation for Micro Aerial Vehicles in Challenging Mars-Analog Environments

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Abstract—In this paper, we present a preliminary performance evaluation of state-of-the-art visual(-inertial) odometries on the INSANE dataset [1]. This dataset contains sequences of an MAV mainly flying through Mars-analog sceneries, equipped with a stereo camera, a monocular camera and an IMU, among other sensors. Challenges arise because of high flying altitudes, fast maneuvers, varying lighting conditions and a homogeneous, unstructured environment. We focus on classical, feature tracking based approaches and evaluate accuracy, robustness and the efficiency of the selected algorithms. With default configurations, VINS-Mono [2] and Granite [3], a modified version of Basalt [4] with mono-VO capabilities, performed best, while VINS-Fusion [5] and OpenVINS [6] failed to complete the two tested Mars-like sequences.

I. INTRODUCTION

Micro Aerial Vehicles (MAVs) offer great possibilities for the exploration of new, unknown terrains. With their ability to quickly fly to points of interest, whose paths may be occluded for rovers and other ground-based robots, they enhance the efficiency and the range of exploring unknown areas. In a terrestrial scope, this benefit is already exploited in search and rescue missions, inspection and exploration tasks. With the first successful flights of the Ingenuity Mars Helicopter in April 2021, it has been proven that the application area of MAVs can be extended towards planetary exploration. Due to long communication times and limited communication windows, a high degree of autonomy is essential for planetary exploration. Furthermore, on Mars one cannot rely on external sensors, like Global Navigation Satellite Systems (GNSS), and navigation algorithms based on visual and inertial sensors have to be used instead. The challenges for such methods are high due to homogeneous and unstructured sceneries on Mars and the fact the MAVs require robust, precise and fast state estimation for stable flights without crashing.

Visual and visual-inertial odometries (VOs and VIOs) have proven their capabilities for this task. Based on data associations between (consecutive) frames, they provide incremental estimation of local poses [7]. For the development of such algorithms, real-world datasets are essential. However, only few datasets combine the challenges imposed by flying systems and unstructured environments and most state-of-the-art algorithms have not been evaluated in this kind of scenario. With the ultimate goal of enhancing the exploration

efficiency and range in the extra-terrestrial context, we want to evaluate a selected amount of state-of-the-art visual(-inertial) odometries on a Mars-analog dataset, recorded by a drone, and identify promising methods and techniques for this specific use-case.

Our main contributions are:

- A first survey on the INSANE dataset, evaluating state-of-the-art visual(-inertial) navigation methods on challenging Mars-analog, drone-recorded, real-world sequences with the experiments running on a Central Processing Unit (CPU) that we also use for real-world missions. To the best of our knowledge, no comprehensive survey on visual(-inertial) odometries has been undertaken on this challenging dataset so far.
- Identification of promising methods or submodules of the methods, that we want to use for further development.
- Sharing lessons learned and challenges during the evaluation of these methods, as a majority of the methods are not designed for Mars-analog environments, and therefore are prone to failure.

This paper is organized as follows: In section II we will give an overview of datasets, that are widely used for evaluating visual and inertial navigation algorithms. In section III we will describe the concrete use-case of our MAV in the context of planetary exploration, introduce the selected dataset and elaborate on the navigation algorithms, that have been chosen for evaluation. In the following section IV we will show and explain the results of the evaluations. In section V we will finally conclude our results, explain our lessons learned and give an outlook for future work.

II. RELATED WORK

For the evaluation and comparison of existing VO and VIO algorithms, a wide range of datasets exist covering all different scenarios and environments. They mostly differ regarding the equipped sensors, the surrounding scenery and the system carrying the sensors. In the scope of this paper we will focus on either MAV-recorded datasets or datasets recorded in homogeneous, unstructured environments. The most-widely used dataset is arguably EUROC [8]. It is recorded with a MAV carrying a stereo camera and an IMU, flying indoors in mostly structured environments. Main challenges are varying

illumination and motion blur in camera images, induced by short, rapid movements. In the UZH-FPV dataset [9] as well as in the UPenn Fast Flight dataset [10] the focus lays on challenging high flight velocities and rapid motion changes. All the mentioned datasets are recorded in mostly structured and inhomogeneous environments. Datasets that focus on outdoor, mostly unstructured and homogeneous sceneries, are for example the MADMAX [11], the Rosario [12], the MARS-LIVG [13] and the INSANE dataset [1]. While Rosario focuses more on agricultural environments, MADMAX as well as INSANE contain sequences that have been recorded in Mars-analog environments. MADMAX has been recorded with a hand-held device, carrying several monocular and stereo cameras as well as IMUs. INSANE has been recorded with a MAV and is equipped with a vast variety of sensors, ranging from multiple IMUs, a stereo and a monocular camera to a laser range finder. Several factors make this dataset very challenging: It includes rapid movements, varying illuminations and homogeneous and unstructured environments. The MARS-LIVG dataset provides outdoor sequences ranging from structured urban sceneries to Mars-analog unstructured environments. A key aspect is that very high flight altitudes for MAVs are covered (up to 130 m above ground).

Currently, no survey exists, comparing VOs and VIOs running on a MAV in the planetary exploration use-case. Delmerico *et al.* have compared different monocular VIO methods in [14]. EUROCC is the hereby selected dataset, which does not contain Mars-analog scenery. Wudenska *et al.* [3] have evaluated monocular VIOs on the MADMAX dataset, lacking the effects of a flying system as the carrier system. Sharafutdinov *et al.* compared modern open-source SLAM approaches, including some VOs and VIOs (e.g. VINS-Mono, Basalt and VINS-Fusion) on a variety of datasets [15]. They include outdoor sceneries, but only in structured urban areas (TUM-VI dataset [16]). Most visual(-inertial) navigation methods, such as VINS-Mono [2], Basalt [4], DeepPatchVO [17] or ORB-SLAM3 [18] evaluate themselves on mostly structured environments such as in EUROCC [8] or TUM-VI [16].

III. EXPERIMENTAL SETUP

In the following paragraph the intended exploration mission and its requirements are described, followed by the preliminary selection of publicly available VO and VIO algorithms. Then, the selected dataset is described, that shall enable an identification of the best suited algorithms and modules for our use case.

A. Mission Description and Requirements

At the Institute of Robotics and Mechatronics (RM) at German Aerospace Center (DLR) a heterogeneous team of robots has been developed as a prototype for exploration missions on other planets or moons [19]. Next to two rovers and a lander, the MAV ARDEA is deployed to enable flexible, efficient and wide-range exploration [20].



Fig. 1. Image of the MAV ARDEA, a system designed for autonomous exploration. Image taken from [20].

ARDEA (shown in figure 1) is equipped with 2 stereo camera pairs, creating 240° vertical Field of View (FOV) and 80° horizontal FOV, and a three-time redundant IMU. A description of the current navigation approach can be found in [21]. The goal is to enhance this approach, in particular the VO, in order to enable long-range flights (> 100 m) at flight altitudes of > 10 m. All processes are run on-board the drone and in real-time. Typical missions can be described in the following manner:

- Take off from a location close to the rovers/lander.
- Travelling to a point of interest at flight altitude > 10 m to overcome small obstacles.
- Collect imagery data or samples.
- Fly back to take-off location to exchange data or samples with other robots.

As of now, all navigation algorithms are run on a Intel NUC13ANHI7 with 32GB of RAM and i7-1360P CPU. The navigation method should not take the majority of available CPU cores, therefore we favor a high efficiency and low computational cost of the methods. Also robustness is a key requirement, as in the planetary exploration context any failure can result in a mission loss. For the most realistic evaluation of the methods regarding our use-case, all tests will be run on an Intel NUC with a real-time constraint.

B. Dataset

We identified the INSANE dataset as the most similar to our use-case. It includes flights in a Mars-analog environment, the flight altitudes range from 3 to 40 m and the velocities are reaching 6 m/s (including rapid and rotation-only movements). As input for visual(-inertial) odometry, the following relevant sensors were used (see [1] for reference):

- **Navigation Cam (mono):** pan-chromatic, 720×540 (downsized from original 2056×1542 to enable real-time feature tracking), global shutter, 20 Hz, downward looking.
- **Mission Cam (stereo):** RealSense T265, pan-chromatic, 848×800 , global shutter, 30 Hz, looking to the front.
- **Pixhawk IMU:** Internally damped, 200Hz.

C. Examined Algorithms

In the scope of this paper only a limited number of VO and VIO algorithms could be examined and further algorithms need to be evaluated in the future, see section V. As of now, we focused on methods that can be run on a modern CPU in real-time and are available as open-source. Furthermore, we focused on methods that use feature tracking as a data association method, like for example the method of Lucas and Kanade [22], because these have proved to work well in the Mars-analog context. This has been demonstrated by the Ingenuity Mars Helicopter [23] and also examined in [3]. The following methods have been selected:

- **VINS-Mono** [2]: A monocular visual-inertial odometry, that optimizes IMU and visual data together in a tightly-coupled way.
- **VINS-Fusion** [5]: An extension of VINS-Mono, resulting in a generalized optimization-based framework, that allows stereo data input.
- **OpenVINS** [6]: A visual-inertial state estimator based on a multi-state constraint Kalman filter [24], allowing for stereo and monocular input.
- **Granite** [3]: Granite is a modified version of Basalt [4] with mono-VO capabilities. In contrast to the approaches mentioned before that rely on OpenCV for feature tracking, Basalt (and thus Granite) uses a self-implemented version of the Lucas-Kanade tracker by the authors of [4].

All methods have not been tuned, they are used in their standard configuration for a fair comparison. Potential options for loop closures (like in VINS-Mono) have been deactivated as we only want to analyse VO and VIO performance. We conducted our own intrinsic and extrinsic calibration based on the calibration data provided and the open-source tool Kalibr [25]. For all methods the Kannala-Brandt camera model was chosen, which allows calibration of cameras with conventional pinhole and fisheye lenses [26]. To the best of our knowledge, this was the only model that all algorithms support as an input. Some of the methods support an online optimization of intrinsic and extrinsic parameters, but these were turned off, to keep computational load low and and the comparison fair. Some of the methods support Multi-Threading (Granite, OpenVINS and VINS-Fusion). In order to find a compromise between computational load and the potential for higher accuracy, the number of threads for data association by means of feature tracking was limited to four. This leaves enough CPU cores for other computations, while also using the enhanced computational performance of multi-threading in the algorithms.

IV. EVALUATION

The accuracy of the selected methods has been evaluated with the Absolute Trajectory Error (ATE) and the Relative Pose Error (RPE), as they are defined in [27]. For VO and VIO systems, those give a good indication of the accumulated drift of a method. The drift needs to be low to ensure that the MAV can return to its base or reach a

waypoint with sufficient accuracy. Both errors have been calculated with the evo library [28]. For computing the ATE the Umeyama SE(3) alignment method [29] (Translation and Rotation) was used. Only for Granite in mono configuration the Sim(3) alignment (Translation, Rotation and Scale) was used, because in this configuration it lacks an absolute scale. For all RPE computations a time intervall Δt of 0.5s is chosen. We average the computations for every configuration over 5 runs on the corresponding sequence.

In order to verify our setup, in particular the calibration methods and the selected camera model, the selected methods were first evaluated on the EUROC dataset and then compared with published benchmarks. Table I shows results for different configurations for the V1-01 sequence. The RSME of the translational ATEs are in the same order of magnitude as reported in previous studies. For example, the translational RMS ATE result of VINS-Mono [2] and OpenVINS [6] are only 0.02-0.03m higher then in their respective papers, which is acceptable keeping in mind that we did not tune the parameters and use a different camera model.

TABLE I
TRANSLATIONAL RMS ATE IN METERS, CALCULATED FOR V1-01
SEQUENCE OF EUROC AND AVERAGED OVER 5 RUNS.

Method	Cams	Imu	RMS ATE
Granite	Stereo	No	0.090
OpenVINS	Stereo	Yes	0.091
OpenVINS	Mono	Yes	0.099
VINS-Mono	Mono	Yes	0.106
VINS-Fusion	Stereo	No	0.514
VINS-Fusion	Stereo	Yes	0.093
VINS-Fusion	Mono	Yes	0.099
Granite	Mono	No	0.096

Evaluations on the Mars-analog sequences of the INSANE dataset were performed in different configurations. Dependent on their capabilities, methods were run as stereo-VO, stereo-VIO, mono-VO or mono-VIO. For monocular configurations the downward looking navigation camera (Nav) or the left camera of the mission cam (SI) was used. The images of the navigation camera had to be downsized to 720×540 pixels. We tested the original resolution of 2056×1542 Pixels, but none of the algorithms could run at this resolution in real-time. As the results of VINS-Mono and Granite will show, the reduced resolution is sufficient. We used the inertial data from the Pixhawk IMU (see [1] for more details on alternative IMUs) because a similar Pixhawk is currently used on our drone ARDEA. A 6DoF ground-truth for these sequences is provided by RTK GNSS and Magnetometer data. The dataset includes a vast variety of trajectories from which we selected a few that fit our use case best. One of them is the Mars10 sequence, which is shown in figure 2, together with the estimated and aligned trajectory of Granite in stereo configuration.

Only few of the selected methods were able to initialize themselves and track the whole trajectory. Table II

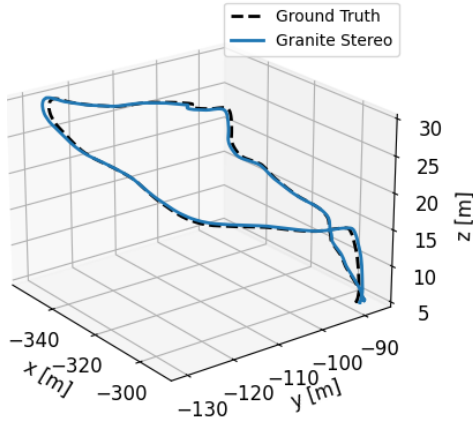


Fig. 2. Trajectory of Granite in stereo configuration on sequence Mars10.

shows the percentage of the trajectory that has successfully been tracked. Only if the whole trajectory was successfully tracked, the error metrics (RMS ATE and RPE) were computed.

TABLE II
PERCENTAGE TRACKED AND THE RMS ATE (TRANSLATIONAL) IN METERS FOR SEQUENCE MARS10, AVERAGED OVER 5 RUNS.

Method	Config	Percentage tracked	RMS ATE
Granite	Stereo	100	1.100
OpenVINS	Stereo+IMU	0	-
OpenVINS	Mono Nav+IMU	0	-
OpenVINS	Mono SI+IMU	0	-
VINS-Mono	Mono Nav+IMU	100	2.509
VINS-Mono	Mono SI+IMU	0	-
VINS-Fusion	Stereo	~ 5	-
VINS-Fusion	Stereo+IMU	~ 5	-
VINS-Fusion	Mono Nav+IMU	0	-
VINS-Fusion	Mono SI+IMU	0	-
Granite	Mono Nav	0	-
Granite	Mono SI	100	1.108

OpenVINS failed to initialize correctly and immediately drifted away in any configuration. This is surprising because the authors of INSANE have demonstrated a successful run of OpenVINS on sequence Mars17 [1]. The setup worked for 2 out of 3 configurations on an indoor sequence of INSANE with a similar calibration file. Therefore, we do not think a faulty calibration is the reason for the malfunction. However, with some parameter tuning, especially regarding the initialization procedure, this method might work.

VINS-Fusion either did not initialize correctly or just managed to track the ascending part of the trajectory (~ 5% of the trajectory), drifting away after reaching flight altitudes of around 10 m. This was unexpected again, because it has shown to work in outdoor environments before. In [30], VINS-Fusion successfully tracked a trajectory, recorded with a drone at a high altitude. However, the scenery was outdoor, but more structured and inhomogeneous than Mars10. Giubileo *et al.* have demonstrated that OpenVINS and VINS-

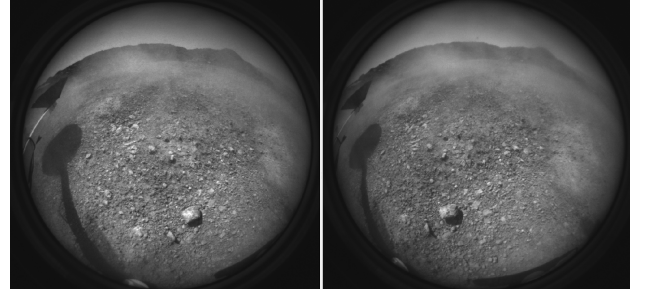


Fig. 3. Recording of the Stereo Camera during takeoff. The downwash of the drone blows up clouds of dust potentially affecting feature tracking.

Fusion can work in a Mars-analog environment on a hand-held sensor setup [31]. VINS-Fusion only worked in 1 out of 4 configurations on the indoor sequences of the INSANE dataset, but in every configuration on the EUROC dataset. We regard the indoor sequences of INSANE as more challenging than EUROC, because of less objects and many untextured areas in the scenery (e.g. homogeneous carpet and white walls), so this is somewhat to be expected. However, a clear reason could not be identified yet. We think that parameter tuning could make the method work here, maybe with the help of the authors, as it was done by Delmerico and Scaramuzza [14].

VINS-Mono performed well in sequence Mars10. At a total distant travelled of 79 m in this sequence, an RMS ATE of 2.5 m equals 3.16%, which is higher than 0.18% in sequence V1-01 of EUROC, but still in a reasonable range.

Granite demonstrated an excellent performance regarding accuracy and robustness. The ratio between travelled distance to RMS ATE is 1.39%, which is also higher than 0.15% in the EUROC dataset, but generally very acceptable with respect to the clearly harsher conditions.

In order to examine higher flight altitudes, all methods were run on sequence Mars11 as well. On this sequence only Granite managed to track the whole trajectory. The conditions during takeoff are very harsh, the downwash of the drone causes a lot of movement in the scenery (see figure 3), potentially leading to an insufficient number of successful features tracked. Also, Granite in stereo configuration managed to track the trajectory relatively accurately, despite the high flying altitudes of almost 40 m. The RMS ATE is 5.901 m. With regard to the distance travelled of 127 m, the ATE makes up 4.65 %, which is a good value, keeping the 4 times higher flight altitudes in comparison to Mars10 and difficult takeoff conditions in mind. It is interesting to note, that the drift is most of the time fairly low, but spikes at specific locations, see figure 4. The spikes may correspond to fast rotation movements (especially at high altitudes), as the comparison to the altitude z and the yaw angle shows. These movements impose a major challenge for MAVs flights, and the visual(-inertial) odometry needs to handle them without orientation loss.

The Mars10 and the Mars11 sequence of the INSANE

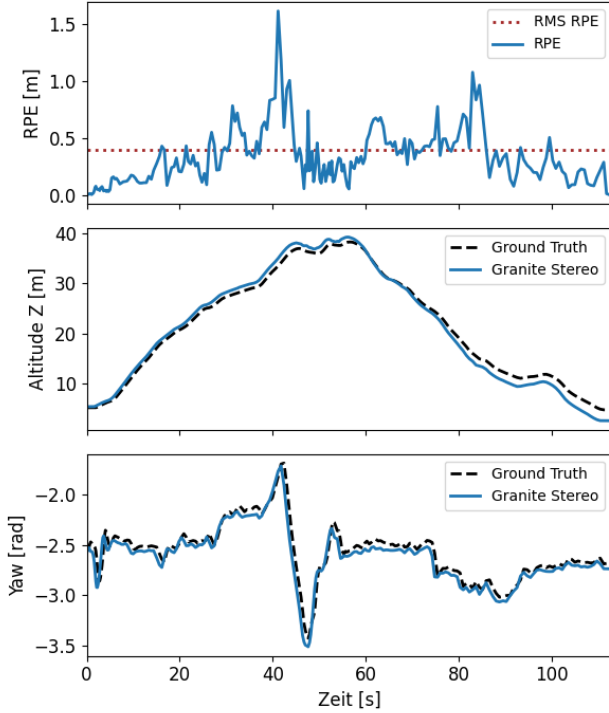


Fig. 4. Computed RPE [m] ($\Delta = 0.5s$) over time and RMS RPE [m] in the Mars11 sequence in comparison to the flight altitude z [m] and yaw angle [rad].

dataset are officially labeled as medium difficulty level (out of easy/medium/difficult) [1]. As many methods failed, we also examined the Mars1 sequence, which is labeled as easy. Here, we saw a similar pattern to Mars10 and Mars11. The initialization procedure was the bottleneck as well, leading to the failure of all methods apart from Granite, which emphasizes the difficulty of this dataset overall.

Regarding computational efficiency, the CPU load percentage has also been tracked. Table III shows that the excellent performance in accuracy and robustness of Granite comes at a higher computational cost. Granite used 5 cores in total (4 for the tracking, 1 for local optimization), all running at almost full load. OpenVINS and VINS-Mono have performed best and especially OpenVINS needs to be highlighted here, because it keeps the CPU load the lowest despite dual camera and inertial input. In general the computational load for the INSANE dataset is higher compared to EUROC, because of higher camera frame rates of the stereo camera and higher resolution of the images.

The most promising results of this paper have shown Granite, VINS-Mono and partially OpenVINS. Granite shows great abilities over long periods of feature tracking, which makes it very precise and robust in the Mars-analog environment. However, as of now, this comes at a high computational cost. VINS-Mono makes a good compromise between accuracy, robustness and efficiency, which makes it a good basis for VIO development for future MAV missions in planetary exploration. Regarding OpenVINS,

TABLE III
COMPUTATIONAL LOAD IN PERCENT, AVERAGED OVER 5 RUNS FOR EACH SEQUENCE.

Method	Config	EUROC V1-01	Insane Indoor	Insane Mars10
Granite	Stereo	15.0	23.0	26.6
OpenVINS	Stereo+IMU	6.8	6.7	-
OpenVINS	Mono Nav+IMU	n/a	-	-
OpenVINS	Mono SI+IMU	5.8	5.7	-
VINS-Mono	Mono Nav+IMU	n/a	8.8	7.6
VINS-Mono	Mono SI+IMU	8.1	11.8	-
VINS-Fusion	Stereo	11.7	-	-
VINS-Fusion	Stereo+IMU	11.8	-	-
VINS-Fusion	Mono Nav+IMU	n/a	-	-
VINS-Fusion	Mono SI+IMU	9.2	12.9	-
Granite	Mono Nav	n/a	13.5	10.6
Granite	Mono SI	12.4	18.8	20.0

the computational efficiency is promising, but in its current configuration it cannot be used yet. However, as others have demonstrated, that it can work in the outdoor/Mars-analog context, we want to enhance the configuration and tune the parameters.

V. CONCLUSION AND FUTURE WORK

In this paper we presented a preliminary evaluation of different state-of-the-art methods in Mars-analog environments. We do not claim that this paper represents a comprehensive survey but regard it as a first step towards the identification of methods that allow robust and accurate state estimation in planetary exploration missions by MAVs. So far, we focused on classical, feature-tracking based approaches. In future work, we want to include other well-established classical approaches like for example ORB-SLAM3 [18], as well as promising, learning-based approaches like for example DeepPatchVO [17]. Learning-based approaches have shown great performance regarding accuracy and robustness, but their capabilities for running on-board in real-time on an MAV need to be explored. Furthermore, we want to see, whether parameters of OpenVINS and VINS-Fusion can be tuned to make them run successfully on the INSANE dataset. Especially, OpenVINS shows great potential regarding efficiency. We identified the initialization process as a major bottleneck for those methods and see the need for parameter tuning especially here.

In this paper, we have already brought methods to their limits, at least in their standard configuration. However, we did not fully exploit all the challenges of the INSANE dataset yet. In the future, we want to examine methods on the most difficult sequences, assuming they were running successfully on the easier ones before. With the future extension of this work, we hope to identify promising methods, that will allow MAVs to safely navigate in planetary exploration missions. Furthermore, after more in-depth analysis, we hope to provide more explanations for the failure of certain methods and give more insight into promising and less promising submodules of the examined methods.

REFERENCES

- [1] C. Brommer, A. Fornasier, M. Scheiber, J. Delaune, R. Brockers, J. Steinbrener, and S. Weiss, “The INSANE dataset: Large number of sensors for challenging UAV flights in Mars analog, outdoor, and out-/indoor transition scenarios,” *The International Journal of Robotics Research*, vol. 43, no. 8, pp. 1083–1113, 2024.
- [2] T. Qin, P. Li, and S. Shen, “VINS-Mono: A robust and versatile monocular visual-inertial state estimator,” *IEEE Transactions on Robotics*, vol. 34, no. 4, pp. 1004–1020, 2018.
- [3] M. Wudenka, M. G. Müller, N. Demmel, A. Wedler, R. Triebel, D. Cremers, and W. Stürzl, “Towards robust monocular visual odometry for flying robots on planetary missions,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021.
- [4] V. Usenko, N. Demmel, D. Schubert, J. Stückler, and D. Cremers, “Visual-inertial mapping with non-linear factor recovery,” *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 422–429, 2019.
- [5] T. Qin, J. Pan, S. Cao, and S. Shen, “A general optimization-based framework for local odometry estimation with multiple sensors,” *arXiv*, 2019.
- [6] P. Geneva, K. Eckenhoff, W. Lee, Y. Yang, and G. Huang, “Open-VINS: A research platform for visual-inertial estimation,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2020.
- [7] D. Scaramuzza and F. Fraundorfer, “Visual odometry [tutorial],” *IEEE Robotics & Automation Magazine*, vol. 18, no. 4, pp. 80–92, 2011.
- [8] M. Burri, J. Nikolic, P. Gohl, T. Schneider, J. Rehder, S. Omari, M. W. Achtelik, and R. Siegwart, “The EuRoC micro aerial vehicle datasets,” *The International Journal of Robotics Research*, vol. 35, no. 10, pp. 1157–1163, 2016.
- [9] J. Delmerico, T. Cieslewski, H. Rebecq, M. Faessler, and D. Scaramuzza, “Are we ready for autonomous drone racing? the UZH-FPV drone racing dataset,” in *International Conference on Robotics and Automation (ICRA)*, 2019, pp. 6713–6719.
- [10] K. Sun, K. Mohta, B. Pfrommer, M. Watterson, S. Liu, Y. Mulgaonkar, C. J. Taylor, and V. Kumar, “Robust stereo visual inertial odometry for fast autonomous flight,” *IEEE Robotics and Automation Letters*, vol. 3, no. 2, pp. 965–972, 2018.
- [11] L. Meyer, M. Smířek, A. Fontan Villacampa, L. Oliva Maza, D. Medina, M. J. Schuster, F. Steidle, M. Vayugundla, M. G. Müller, B. Rebele, A. Wedler, and R. Triebel, “The MADMAX data set for visual-inertial rover navigation on Mars,” *Journal of Field Robotics*, vol. 38, no. 6, pp. 833–853, 2021.
- [12] T. Pire, M. Mujica, J. Civera, and E. Kofman, “The Rosario dataset: Multisensor data for localization and mapping in agricultural environments,” *The International Journal of Robotics Research*, vol. 38, no. 6, pp. 633–641, 2019.
- [13] H. Li, Y. Zou, N. Chen, J. Lin, X. Liu, W. Xu, C. Zheng, R. Li, D. He, F. Kong, Y. Cai, Z. Liu, S. Zhou, K. Xue, and F. Zhang, “MARS-LVIG dataset: A multi-sensor aerial robots SLAM dataset for LiDAR-visual-inertial-GNSS fusion,” *The International Journal of Robotics Research*, vol. 43, no. 8, pp. 1114–1127, 2024.
- [14] J. Delmerico and D. Scaramuzza, “A benchmark comparison of monocular visual-inertial odometry algorithms for flying robots,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2018.
- [15] D. Sharafutdinov, M. Griguletskii, P. Kopanev, M. Kurenkov, G. Ferrer, A. Burkov, A. Gonnochenko, and D. Tsetserukou, “Comparison of modern open-source visual SLAM approaches,” *Journal of Intelligent & Robotic Systems*, vol. 107, no. 3, 2023.
- [16] D. Schubert, T. Goll, N. Demmel, V. Usenko, J. Stückler, and D. Cremers, “The TUM VI benchmark for evaluating visual-inertial odometry,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2018.
- [17] Z. Teed, L. Lipson, and J. Deng, “Deep patch visual odometry,” *arXiv*, 2022.
- [18] C. Campos, R. Elvira, J. J. G. Rodriguez, J. M. M. Montiel, and J. D. Tardos, “ORB-SLAM3: An accurate open-source library for visual, visual-inertial, and multimap SLAM,” *IEEE Transactions on Robotics*, vol. 37, no. 6, pp. 1874–1890, 2021.
- [19] M. J. Schuster, M. G. Müller, S. G. Brunner, H. Lehner, P. Lehner, R. Sakagami, A. Domel, L. Meyer, B. Vodermayr, R. Giubilato, M. Vayugundla, J. Reill, F. Steidle, I. von Barga, K. Bussmann, R. Belder, P. Lutz, W. Stürzl, M. Smisek, M. Maier, S. Stoneman, A. F. Prince, B. Rebele, M. Durner, E. Staudinger, S. Zhang, R. Pohlmann, E. Bischoff, C. Braun, S. Schroder, E. Dietz, S. Frohmann, A. Börner, H.-W. Hubers, B. Foing, R. Triebel, A. O. Albu-Schaffer, and A. Wedler, “The ARCHES space-analogue demonstration mission: Towards heterogeneous teams of autonomous robots for collaborative scientific sampling in planetary exploration,” *IEEE Robotics and Automation Letters*, vol. 5, no. 4, pp. 5315–5322, 2020.
- [20] P. Lutz, M. G. Müller, M. Maier, S. Stoneman, T. Tomić, I. von Barga, M. J. Schuster, F. Steidle, A. Wedler, W. Stürzl, and R. Triebel, “ARDEA—an MAV with skills for future planetary missions,” *Journal of Field Robotics*, vol. 37, no. 4, pp. 515–551, 2020.
- [21] M. G. Müller, F. Steidle, M. J. Schuster, P. Lutz, M. Maier, S. Stoneman, T. Tomić, and W. Stürzl, “Robust visual-inertial state estimation with multiple odometries and efficient mapping on an MAV with ultra-wide FOV stereo vision,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2018.
- [22] B. D. Lucas and T. Kanade, “An iterative image registration technique with an application to stereo vision,” *International Joint Conference on Artificial Intelligence (IJCAI)*, 1981.
- [23] D. S. Bayard, D. T. Conway, R. Brockers, J. H. Delaune, L. H. Matthies, H. F. Grip, G. B. Merewether, T. L. Brown, and A. M. San Martin, “Vision-based navigation for the NASA Mars helicopter,” in *AIAA Scitech Forum*, 2019.
- [24] A. I. Mourikis and S. I. Roumeliotis, “A multi-state constraint Kalman filter for vision-aided inertial navigation,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2007.
- [25] P. Furgale, J. Rehder, and R. Siegwart, “Unified temporal and spatial calibration for multi-sensor systems,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2013.
- [26] J. Kannala and S. Brandt, “A generic camera model and calibration method for conventional, wide-angle, and fish-eye lenses,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 28, no. 8, pp. 1335–1340, 2006.
- [27] J. Sturm, N. Engelhard, F. Endres, W. Burgard, and D. Cremers, “A benchmark for the evaluation of RGB-D SLAM systems,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2012, pp. 573–580.
- [28] M. Grupp, “evo: Python package for the evaluation of odometry and SLAM,” <https://github.com/MichaelGrupp/evo>, 2017.
- [29] S. Umeyama, “Least-squares estimation of transformation parameters between two point patterns,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 13, no. 4, pp. 376–380, 1991.
- [30] A. George, N. Koivumäki, T. Hakala, J. Suomalainen, and E. Honkavaara, “Visual-inertial odometry using high flying altitude drone datasets,” *Drones*, vol. 7, no. 1, p. 36, 2023.
- [31] R. Giubilato, W. Stürzl, A. Wedler, and R. Triebel, “Challenges of SLAM in extremely unstructured environments: The DLR planetary stereo, solid-state LiDAR, inertial dataset,” *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 8721–8728, 2022.